

SAMPLE final written test – Basics in linear algebra and numerical methods, October 2026

Here are some questions so you can practice for the final exam, which will contain questions similar to this (in fact the format is the same). Each questions is either multiple choice (where sometimes you have to pick more than one option) or just write the answer in 1 sentence. The final exam will have **19 questions**, each worth the same amount of points for a total of 47.5 points. However, there will be a couple of extra questions, which will be worth extra points in case you missed a homework or deadline or you just want to challenge yourself. **Notes are allowed (preferably printed notes) and the use of internet, or python, is forbidden.**

- For which kind of matrices or vectors A does AA^T make sense?
(a) square matrices (b) square matrices and row vectors
(c) square matrices and any vectors (d) any matrices and vectors (e) none of the above
- Let $A \in \mathbb{R}^{n \times m}$ such that $n > m$. Suppose that there exists $b \in \mathbb{R}^n$ such that $Ax = b$ has a solution. Which of the following is not necessarily true:
(a) The solution is unique (b) $b \in \text{Col}(A)$ (c) $\text{Rank}(A) \leq m$ (d) $\dim(\text{Null}(A)) = 0$.
- Let $A = \begin{pmatrix} 3 & 1 & 3 \\ 0 & 4 & -1 \\ -3 & 4 & 5 \end{pmatrix}$. Suppose you want to solve $Ax = b$ using the Jacobi method. Which of the following relative error sequences is compatible with the method's known behavior?
(a) $[1, 2, 3, 4]$ (b) $[1, 0.5, 0.25, 0.125]$ (c) $[1, 1, 1, 1]$ (d) $[1, 10^{-1}, 10^{-3}, 10^{-6}]$
- In the previous exercise. What do we expect the residual error to be if we apply the least square method? Why?
- Which of the following matrices correspond to the mirroring to the line $x = y$ in $\mathbb{R}^2$?
(a) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ (b) $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ (c) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (d) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$
- Someone multiplies the full matrices $A, B \in \mathbb{R}^{N \times N}$ without making any tricks and extra observations. How many multiplications has to be computed for this?
(a) N^2 (b) $2N^2$ (c) N^3 (d) $2N^3$ (e) more
- The sparse matrix $S \in \mathbb{R}^{500 \times 500}$ contains 10000 non-zero elements and this is multiplied with a column vector of size 500 (all non-zeros). How many multiplications (of real numbers) have to be performed then?
(a) It depends on the structure of S (b) $10000 \cdot 500$ (c) 10000 (d) $500 \cdot 500$
- Take an array $A \in \mathbb{R}^{3 \times 3 \times 3}$ and consider
 $B_1 = A[0, :, :] + A[1, :, :] + A[2, :, :]$ and $C_1 = B_1[0, :] + B_1[1, :] + B_1[2, :]$.
Similarly, we take
 $B_2 = A[:, 0, :] + A[:, 1, :] + A[:, 2, :]$ and $C_2 = B_2[0, :] + B_2[1, :] + B_2[2, :]$.
Which of the following statements are (is) true?

- (i) C_1 has dimension 3×3 .
(ii) $C_1 = C_2$ in all cases.
(iii) The sum of the elements in B_1 and B_2 are equal.
- (a) (ii) and (iii) (b) (i) and (iii) (c) only (ii) (d) only (iii) (e) none of them
9. Let $P \in \mathbb{R}^{n \times 2}$. What is the size $P[\text{None}, :, :] - P[:, \text{None}, :]$?
(a) (1,n,2) (b) (n,n,2) (c) (1,n,n) (d) (n,2,2)
10. Let us interpret the rows in the array P in the previous question as the x, y coordinates of different points in $\mathbb{R}^2$. In just one line, how would you compute the Euclidean distance matrix of these points.
11. Let P be the permutation matrix corresponding to the permutation $[1, 2, 3, 4, 5, 6, 7, 8] \rightarrow [3, 4, 6, 1, 2, 5, 7, 8]$. Find the smallest $n \in \mathbb{N}$ such that $P^n = I$, where $I \in \mathbb{R}^{8 \times 8}$ denotes the identity matrix.
12. We know that $(\mathbf{v}, \lambda_1)$ is an eigenvector-eigenvalue pair for the matrix $A_1 \in \mathbb{R}^{N \times N}$, and similarly, $(\mathbf{v}, \lambda_2)$ is an eigenvector-eigenvalue pair for the matrix $A_2 \in \mathbb{R}^{N \times N}$. We can state then for sure that ... is an eigenvector-eigenvalue pair for the matrix $A_1 + A_2 \in \mathbb{R}^{N \times N}$.
(a) $(\mathbf{v}, \lambda_1 + \lambda_2)$ (b) $(2\mathbf{v}, \lambda_1 + \lambda_2)$ (c) $(\mathbf{v}, \lambda)$ (d) $(\mathbf{v}, \lambda^{-3})$ (e) none of them
13. Suppose A and B are similar matrices, i.e., $B = S^{-1}AS$ for some invertible matrix S . Let Which of the following statements is not necessarily true:
(a) $\det(A - \lambda I) = \det(B - \lambda I)$ (b) $B^n = S^{-1}A^nS$ (c) $A = SBS^{-1}$ (d) $S = S^T$
14. We define the relation $\sim$ on the set of matrices $A \in \mathbb{R}^{N \times N}$ as follows: $A \sim B$ if and only if there is a permutation matrix $P \in \mathbb{R}^{N \times N}$ so that $A = PB$. Which of the properties
(i) reflexive (ii) symmetric (iii) transitive
is true for the relation $\sim$?
(a) (i) and (ii) (b) (i) and (iii) (c) (ii) and (iii) (d) all of them (e) only (i)
15. We take a matrix $A \in \mathbb{R}^{4 \times 3}$ and change two of its columns by some arbitrary vectors of size 4. Then its rank may also be changed by
(a) 0 or 1 (b) 0 or 1 or 2 (c) 0 or 1 or 2 or 3 (d) 1 or 2 (e) 0 or 2
16. The rank of matrices $A, B \in \mathbb{R}^{4 \times 4}$ are both 2. What can be the rank of the matrix AB ?
(a) 0 or 1 (b) 0 or 1 or 2 (c) 0 or 1 or 2 or 3 (d) 0 or 2 (e) none of them
17. How many of the following functions ρ define a *metric* on the set of binary vectors of length N ?
 $\rho(\mathbf{u}, \mathbf{v}) = \sum_{j=1}^N |u_j - v_j|$ $\rho(\mathbf{u}, \mathbf{v}) = \sum_{j=1}^{N-2} u_j - v_j$ $\rho(\mathbf{u}, \mathbf{v}) = \max u_j - v_j$ $\rho(\mathbf{u}, \mathbf{v}) = \left(\sum_{j=1}^N |u_j - v_j| \right)^2$
(a) 0 (b) 1 (c) 2 (d) 3 (e) 4

18. How many of the following formulas applied to the vector $[a_1, a_2, \dots, a_N]^T$ define a *norm* on $\mathbb{R}^N$?

$$\sum_{j=1}^N |a_j| \quad \sum_{j=1}^N \sqrt{a_j} \quad \max_j a_j^2 \quad \min |a_j|$$

- (a) 0 (b) 1 (c) 2 (d) 3 (e) 4

19. For the matrix $A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and the vector $\mathbf{v} = [7, 8, 9]^T$ we perform the algorithm $\mathbf{v}^{n+1} = A\mathbf{v}^n$.
What is the limit of $\frac{\|A\mathbf{v}^n\|}{\|\mathbf{v}^n\|}$?

- (a) 1 (b) 2 (c) 3 (d) 0 (e) none of them

20. What is the size of the matrices U, V and D in the reduced SVD decomposition of a matrix $A \in \mathbb{R}^{4 \times 100}$, if we retain all singular values?

21. In the gradient method, we want to minimize the function $f(x, y, z) = x - zx + 3(y - 3)^2 z$ by starting from $(x_0, y_0, z_0) = (1, 2, 1)$ using $\alpha = 0.1$. What is the next value (x_0, y_0, z_0) ?

- (a) (0.9, 2.1, 1.2) (b) (1.2, 1.6, 1.2) (c) (1.1, 1.5, 1.2) (d) (1.2, 2, 1.1) (e) none of them

22. We are looking for the matrices $A_1, A_2 \in \mathbb{R}^{40 \times 40}$ and the vector $\mathbf{b} \in \mathbb{R}^{40}$ such that for a given $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{40}$ vectors $|\mathbf{y} - A_2(A_1\mathbf{x} - \mathbf{b})|^2$ is minimal. Using the gradient method what is the size of the gradient vector to work with?

- (a) 1600 (b) 1640 (c) 3200 (d) 3280 (e) none of them